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Contrasting effects of compositional shifts and resource fluctuations on phytoplankton nutrient uptake under warming

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Keywords: phytoplankton, warming, nutrient load, nutrient uptake, minimum resource requirement

Publication details:

Károly Pálffy (2026): Contrasting effects of compositional shifts and resource fluctuations on phytoplankton nutrient uptake under warming. *Journal of Plankton Research*, 48(6), fbag084, https://doi.org/10.1093/plankt/fbag084

Abstract

Understanding how global warming-induced changes in phytoplankton affect nutrient cycling is essential for assessing climate change impacts in freshwater environments. The present study used a multispecies model to explore the effect of temperature on nutrient uptake efficiency and its link with community structure and resource use-related traits. Simulations were run under four different annual temperature patterns (baseline to +3°C), along various nutrient loads representing a nitrogen-to-phosphorus (N:P) ratio gradient, and under constant and fluctuating supply. Warming led to more balanced annual nutrient uptake and increasing mean uptake efficiency. Under constant loads, this increase depended on the N:P ratio and on compositional shifts. In communities exhibiting high species turnover under warming, increase in nutrient uptake was constrained by higher minimum resource requirements (R^*) during the nutrient-limited summer period. Higher fluctuations in nutrient loads enhanced warming-induced increases in uptake efficiency, independently of shifts in R^*, also increasing annual net growth time. These results suggest that warming impacts on nutrient uptake can be traced back to multiple mechanisms: trait-mediated changes in competitive ability, and changes in the length of the growth period. Thus, the relative importance of community reassembly and environmental variability might have substantial influence on phytoplankton-mediated ecosystem responses to climate change.

Introduction

As the consequences of global warming are becoming increasingly apparent, studies focusing on this pressing issue are concurrently mushrooming in scientific literature. The effects of climate change on ecosystems are highly complex, making the assessment of its putative ecological consequences a complicated task (Turner et al., 2020). A key question regarding the long-term impacts of warming is how elemental cycles might be altered, and in what ways these alterations interact with the composition and functioning of communities (Jarvie et al., 2012). In freshwater environments, warming-induced effects on nutrient cycling can also lead to water quality-related issues, affecting both ecological status and human use (Woolway et al., 2022). These tendencies can have important consequences in lakes of the temperate zone, where climate change impacts have been shown to be particularly pronounced (Piccolroaz et al., 2020).

Elucidating long-term changes in nutrient cycling necessitates the compartmentalized study of nutrient budget components, otherwise observed trends cannot be interpreted and explained in terms of the underlying processes. As a major primary producer, phytoplankton has a crucial role in this regard, but its dynamics are shaped by species-specific growth and nutrient uptake kinetics, whose complex dependence on numerous global change stressors necessitates more in-depth research (Van de Waal & Litchman, 2020). Understanding these interrelationships is currently limited, as relevant data on species-level responses along broad environmental gradients is sporadic (Bestion et al., 2018; Levasseur et al., 2025; Lewington-Pearce et al., 2019). This knowledge gap, however, does not fully restrain efforts to test hypotheses on phytoplankton dynamics and changes in community functioning under a warming climate (Kremer et al., 2025). Process-based models that apply thermal growth curves to simulate algal community responses under various temperature scenarios (Grimaud et al., 2017) are widely used to explore mechanistic links between changing climate variables and phytoplankton-related ecosystem processes.

Several modelling studies are aimed at system stability, exploring the effect of environmental gradients and community structure on equilibrium conditions through long-term simulations (e.g. Ceulemans et al., 2019; Cowall et al., 2021; Huisman & Weissing, 1999). In contrast with this approach and in accordance with numerous ecological systems exhibiting transient dynamics (Hasting et al., 2018), some process-based modelling works on phytoplankton focus on short-term simulations (Jäger et al., 2008). Since climate change has a substantial effect on seasonal phytoplankton dynamics (Ardyna & Arrigo, 2020; De Senerpont Domis et al., 2013; Nõges et al., 2010), narrowing down the simulation period to the annual cycle may indeed carry more ecological relevance.

Regarding bottom-up effects, alterations in phytoplankton community structure trace back to intricate relationships between temperature-dependent, species-specific growth rates and resource supply. Investigating the nature of these relationships also requires an appropriate choice of relevant traits. The minimum resource requirement at which species growth and mortality is in balance (R^*; Tilman, 1982) is a widely used, temperature-dependent measure in competition theory, which can be a powerful tool to predict competitive outcomes in phytoplankton communities (Gallego & Narwani, 2022). In addition to competition, this feature of the response curve also indicates to what level a species is able to draw down a specific nutrient under limiting conditions, which makes it an appropriate candidate to connect temporal dynamics and nutrient uptake efficiency. Nevertheless, the role of temperature-dependent resource acquisition traits on the community level under continuously changing conditions over the annual cycle remains largely unclear. Seasonality is a key factor in this regard, as in most temperate lakes low temperature is a strong limiting factor in winter, while summer is generally the period of nutrient limitation (Sommer et al., 2012). Consequently, the seasonal shift in the relative effects of temperature and nutrient availability on species-specific growth rates is a feature that can be inherently altered under a warming climate. In addition to growth, the seasonality impact is also manifested in how the importance of various traits changes over the year, with nutrient utilization traits generally becoming vital in the nutrient-limited summer period (Edwards et al., 2013). This leads to the assumption that long-term warming can induce changes in seasonal trait significance, expanding the period when nutrient use-related traits are crucial, which thus might also affect the aggregate nutrient uptake of the phytoplankton. A similarly intriguing question is whether elevated mean temperatures, by facilitating the dominance of species with higher thermal optima, can ultimately modify uptake efficiency through the communities' concurrently altered trait structure.

In a recent paper, we used a multispecies, multi-nutrient dynamic model to investigate the effect of elevated mean temperature on the seasonal variability of phytoplankton composition (Pálffy & Smeti, 2024), based on a long-term observation in a large, shallow, temperate lake (Pálffy & Vörös, 2019). The present paper relies on the same model, re-analyzing the simulation outputs from a functioning-oriented aspect, focusing on the impact of warming and various nutrient supply regimes on phytoplankton compositional shifts, minimum resource requirements and nutrient uptake. The initial hypothesis of the study was that increasing mean temperature induces higher total nutrient uptake efficiency over the annual cycle, which positively correlates with the rate of compositional shift and negatively with community-weighted minimum resource requirements. Due to the anticipated impact on seasonal trait significance, shifts during the nutrient-limited period were given careful consideration.

Method

Model structure

The applied model describes the temporal dynamics of phytoplankton communities exposed to various nitrogen (N) and phosphorus (P) loads, and to different annual temperature patterns (Fig. 1.). The model code, simulation and analysis scripts, and species-specific model parameters are available in a public GitHub repository (https://github.com/k-palffy/phyto_warming_model). The approach was based on the model of Thomas et. al (2017), formulating the dependence of algal growth rate on temperature and nutrient concentration. To facilitate simulation on the community level, this model was embedded into a multispecies, multi-nutrient model (Roelke & Spatharis, 2015). Thus, the model calculated species-specific growth rates as follows:

$$\mu_{ij}(T, R_j) = b_{1ij} \cdot exp(b_{2ij} \cdot T) \cdot R_j/(R_j + K_{ij}) - (d_{0ij} + d_{1ij} \cdot exp(d_{2ij} \cdot T)) \quad (1)$$

where μ_{ij} is the growth rate of species i as a function of temperature (T) and resource j (N or P). All variables and model parameters can be found in Table I. The multispecies, multi-nutrient model represented a community in a flow-through system. Instantaneous species-specific growth rates were determined based on Liebig's law of the minimum and Equation 1 as:

$$\mu_i = min[\mu_i(T, N), \mu_i(T, P)] \quad (2)$$

Temporal changes in species abundances and nutrient concentrations were calculated after determining instantaneous growth rates from Equations 1 and 2 using the following equations:

$$dA_i/dt = \mu_i \cdot A_i - v \cdot A_i \quad (3)$$

$$dR_j/dt = v \cdot (R_{Sj} - R_j) - \sum_{i=1}^{n}(q_{ij} \cdot \mu_i \cdot A_i) \quad (4)$$

All the tasks, including the definition of the model structure, simulations and data analysis were performed using the R programming language version 4.4.1 (R Core Team, 2024).

Simulations

Species-specific model parameters were determined with maximum likelihood estimation using the *bbmle* package (Bolker and R Development Core Team 2021) from literature data (Lewington-Pearce et al., 2019; Mastala et al., 1996; Amano et al., 2010; Takeya et al., 2012; Thomas and

Litchman, 2016; Bestion et al., 2018; Cho et al. 2018). Robust estimation of the parameters in the model of Thomas et al. (2017) requires broad thermal and nutrient gradients, thus, only papers with a broad coverage of temperature and N or P treatments were suitable for parameterization. Studies that met this requirement contained data on three species of chlorophytes (*Scenedesmus*, *Pediastrum*, *Raphidocelis*), two species of cyanobacteria (*Microcystis* and *Synechococcus*) and one diatom species (*Cyclotella*). In cases where data availability allowed, identical species data were used for N and P, while in cases where data for the same species were unavailable, studies on a species of the same genus served as data source. Estimation of the model parameters for N and P was performed separately, since relevant papers contained net growth rates along temperature and only one nutrient at a time. This also necessitated that in addition to the half-saturation constant (K), separate sets of values were estimated for both the growth (b_1, b_2) and mortality terms (d_0, d_1, d_2) in the case of N and P, respectively. As a parameter matrix of six species overly constrains community assembly and can lead to limited simulation outcomes, further parameter combinations were added by shifting the original literature data on growth rates along the gradients of temperature and nutrients. The model was refitted on these modified data using the same method, expanding the original matrix of six to 90 species. Phytoplankton was thus represented by communities of 10, 30 or 50 species randomly assembled from this expanded matrix containing both real and 'virtual', nonetheless realistic species parameters. The randomization loop was set to generate one hundred unique communities for each species richness level, resulting in a total of 300 different virtual communities (Fig. 1). The initial abundance of each species was 1000 cells mL^{-1}, and each simulation was run for a simulation period of two years, where the second year was used for analysis to avoid the effect of initial conditions in the first year.

The aim of the original modelling approach (Pálffy & Smeti, 2024) was to test a hypothesis regarding long-term changes in Lake Balaton, a large shallow lake in Central Europe. Therefore, the simulation environment was adjusted to represent locally characteristic abiotic conditions. Annual temperature pattern characteristic of Lake Balaton served as a baseline scenario. This annual pattern was acquired from data provided by the Hungarian Meteorological Service, covering years between 1982 and 2006. Daily values were estimated by fitting a generalized additive mixed model (GAMM) with an autocorrelation structure of order 1 (AR-1) on the available data series (Zuur et al., 2009; Pálffy & Vörös, 2019), and year 2000 was used as the baseline period. Simulations representing warming were generated by increasing each daily value of the baseline scenario by 1°C, 2°C or 3°C, yielding four scenarios of identical temporal pattern with different annual minimum, maximum and mean values (Fig. 1).

Lake Balaton has one major inflow, entering the lake at its westernmost point, therefore simulated nutrient loads and hydrological discharge were set to represent ranges characteristic of

the inflow and the directly affected western basin of the lake (Tátrai et al. 2000). As daily changes in nutrient concentrations were simulated using hydraulic flushing rate (v) and concentrations in the source (R_{Sj}; Equation 4), the values of these parameters needed to be derived from the available data. Accordingly, v was obtained by dividing daily hydrological discharge with the estimated average water volume of the western basin, and R_{Sj} was computed by dividing daily nutrient loads with the daily discharge. Simulated nutrient loads followed one of two basic temporal patterns: constant or fluctuating load and discharge (Table II). Constant conditions consisted of the combination of three different N and P load levels (i.e. a total of nine combinations), and a hydrological load of 685,000 m^3 d^{-1}. Fluctuating conditions meant three different levels of temporal variation: nutrient and hydrological loads changing every 30, 20 or 10 days. Within each fluctuating environment, load values were randomly altered within ranges representing three different nutrient levels. Hydrological load was set to change between 548,000 and 822,000 m^3 d^{-1} within each variation-nutrient load combination. To increase robustness of the simulation outputs, each combination was re-generated ten times with randomly modified load values. In total, 4 (temperature scenarios) × 9 (nutrient load combinations) × 300 (communities) = 10,800 simulations were run with constant loads, while the number of fluctuating load simulations added up to 4 (temperature scenarios) × 90 (nutrient load combinations) × 300 (communities) = 108,000. Model equations were solved using fifth-order Runge–Kutta method with Dormand–Prince coefficients (Dormand & Prince, 1980), variable time step, and minimum integration step size set to 0.01 (day). Each simulation yielded an output data frame containing daily values of the abundance of each species of the community and of the concentrations of N and P.

Data analysis

Since each species was represented with different cellular nutrient quota, which inherently correlates with cell volume, all simulated species abundance values were converted into biovolume prior to analysis. Cell volumes for the six original species were determined using average values from literature, while for the virtual species, cell volumes were modified with the same species-specific random multipliers as those used for cellular nutrient quota. To link phytoplankton community structure with functioning, simulations were assessed using three measures: annual nutrient uptake efficiency (η_j), biovolume-weighted minimum resource requirements (R^*_j), and the number of days when mean growth rate was positive (hereafter termed net growth time, t_g). Nutrient uptake (U_j) was determined for each simulation day (t) using the following equation:

$$U_{jt} = v_t \cdot \left(R_{Sjt} - R_{jt}\right) - \Delta R_{t,t+1} \qquad (5)$$

Nutrient uptake efficiency was computed as the ratio of daily nutrient uptake values summed over the year and total annual nutrient load:

$$\eta_j = \frac{\sum U_{jt}}{(\sum v_t \cdot R_{Sjt})} \qquad (6)$$

Species-specific daily values of minimum resource requirement (R^*_{ij}) were determined for each simulation step and subsequently weighted with the relative biovolume of the species to obtain daily community-wide means. The minimum resource requirement was determined using Equation 1 by taking instantaneous growth rate (μ_{ij}) as zero. After rearranging the equation, we get the following formula:

$$R^*_{ij} = \frac{K_{ij} \cdot (d_{0ij} + d_{1ij} \cdot exp(d_{2ij} \cdot T))}{b_{1ij} \cdot exp(b_{2ij} \cdot T) - (d_{0ij} + d_{1ij} \cdot exp(d_{2ij} \cdot T))} \qquad (7)$$

Daily species-specific growth rates were determined using Equations 1 and 2, biovolume-weighted means were computed the same way as for R^*_{ij}. Net growth time (t_g) was determined by counting the number of days when biovolume-weighted means were positive. Species-specific thermal optima (i.e. the value at which growth rate reaches its maximum along the temperature gradient) were also determined for each simulation day with function *optimize()* of the *stats* package (R Core Team, 2024), using Equations 1-2 and daily nutrient concentrations. To assess the rate of warming-induced compositional shifts, relative biovolume-based Bray-Curtis dissimilarities were computed between warming scenario simulations and their specific baseline scenario counterpart for each day of the simulation outputs.

Annual patterns of total biovolume, inorganic nitrogen and phosphorus concentration and uptake in various temperature-nutrient load combinations were compared by fitting generalized additive mixed models (GAMMs) onto the daily simulation outputs. Separate GAMMs were fitted for each temperature-nutrient load combination and initial species richness level, i.e. each GAMM fit was based on data from 100 unique simulations. An autocorrelation structure of order one (AR-1) was used to correct for temporal dependence between subsequent daily values. To understand the influence of various nutrient load levels and fluctuations, a principal component analysis (PCA) was performed on changes in aggregated annual values of nitrogen and phosphorus uptake efficiency (η_N, η_P), net growth time (t_g), minimum nitrogen and phosphorus requirement (R^*_N, R^*_P) and Bray-Curtis dissimilarity. The data was filtered for simulations run in the +3°C scenario with an initial

species richness of 50 to increase clarity in the ordination plot. Thus, the PCA included the pairwise differences in annual values and compositional dissimilarities between the baseline and +3°C simulations.

Finally, generalized least squares models (GLS) and GAMMs were fitted onto η_N and η_P as response variables to test the effects of temperature, N/P ratio, initial species richness, R^*_N and R^*_P, t_g, and fluctuations. The constant load simulations allowed for the assessment of pure seasonality effects, therefore the values of uptake efficiency (η_N, η_P) and minimum resource requirement (R^*_N, R^*_P) were computed for the whole year as well as for the period of nutrient limitation. The period of nutrient limitation was determined for each simulation as the time frame within which daily nutrient concentration (R_N or R_P) was below 110% of daily biovolume-weighted mean R^*_N or R^*_P, respectively. All explanatory variables were standardized to have zero mean and unit variance prior to model fitting to minimize scale differences. GAMMs were chosen for the constant load simulations, because η_N and η_P showed nonlinear changes along N/P. GLSs were sufficient for fluctuating load simulations, where each fluctuation and load level was allowed to vary around the Redfield ratio, thus, the effect of N/P was not relevant in the analysis. For both GLSs and GAMMs, multiple models were fitted with different variance structures, and the best fit was chosen based on Akaike's information criterion (AIC) (Sakamoto et al., 1986). Preliminary data visualization showed that in fluctuating simulations fluctuation as a condition *per se* had a substantially stronger impact than its frequency, thus, it was included as a binary (presence/absence) variable in model fitting. Furthermore, to moderate model complexity and to make the joint interpretation of model and PCA outputs feasible, in the case of fluctuating simulations only communities with an initial species richness of 50 were included.

Results

In constant load simulations, the impact of elevated mean temperature on total phytoplankton biovolume, nutrient assimilation and community composition was strongly influenced by the N:P load ratio (Fig. 2). As expected, significant temperature-induced changes in biovolume mostly occurred in winter-early spring and late autumn-winter periods in all constant load combinations. Annual minimum values of total biovolume were higher under elevated temperatures, also leading to lower maximum inorganic nutrient concentrations. Particularly at N:P ratios close to the Redfield ratio (16:1), higher mean temperature mitigated seasonal fluctuations in total biovolume. This mitigating effect was also manifested in daily nutrient uptake, i.e., warming scenarios allowed smaller temporal fluctuations in uptake as compared to the baseline scenario (Fig. 2). Daily Bray-Curtis dissimilarities from the baseline scenario showed different patterns along the N:P load ratio.

Below the Redfield ratio, warming-induced dissimilarity was low and increased only around the winter period, whereas close to and above the ratio, dissimilarity was considerably higher throughout the whole simulation period (Fig. 2). The GAMM fits of biovolume-weighted mean growth rate yielded expected patterns, elevated temperature scenarios caused broader net growth periods (Fig. S2). Although the N/P load ratio influenced the impact of warming on the weighted mean thermal optimum (T_{opt}), summer periods induced higher T_{opt} in the higher temperature scenarios in comparison with the baseline (Fig. S2).

Annual nutrient uptake efficiency changed in particular patterns along the N:P ratio, showing higher values for the limiting nutrient: high η_N switched to high η_P above the Redfield ratio (Fig. 3). This change was also visible in the warming effect, higher mean temperatures led to higher increase in the uptake efficiency of the limiting nutrient. Besides the N:P ratio, this warming-induced change in uptake efficiency also correlated with the rate of compositional shift. In those warming simulations where annual mean dissimilarity from the baseline scenario was higher, the increase in annual nutrient uptake efficiency was smaller (Fig 4). In contrast with this decline, annual mean of biovolume-weighted T_{opt} showed a warming-induced increase when dissimilarity from the baseline scenario was high. Although t_g did not show a similarly monotonic trend, its highest warming-induced increase also occurred in high-dissimilarity simulations. Best GAMM fits showed similar relationships (Table III), N/P had a significant non-linear effect on η_N and η_P and a significant tensor product interaction with temperature scenarios. Temperature *per se*, initial species richness and t_g showed significant positive effects for both nutrients, whereas R^* had a weak positive impact in the case of N and a negative effect for P (Fig. S1).

When the assessment was narrowed down to the summer period of nutrient limitation, the correlation between R^* and uptake efficiency became more apparent. In warming simulations where mean dissimilarity from the baseline scenario was high, the relative increase in mean R^* was accompanied by a relative decrease in uptake efficiency (Fig. 5, Pearson's $r = -0.659$, $p < 0.0001$), and changes in mean T_{opt} showed a similar relationship (Fig. S3).

When nutrient load was allowed to fluctuate, warming-induced changes in annual nutrient uptake efficiency showed a different pattern (Fig. 6). In the case of low nutrient supplies, fluctuations led to an overall decrease in annual uptake efficiency compared to constant load simulations, although random fluctuations also enhanced variation in the output. On the other hand, increases in uptake efficiency at high and fluctuating nutrient load level surpassed those under constant load conditions. The frequency of the fluctuations had a weaker impact on uptake efficiency than fluctuation *per se*, but there was a gradual increase in variance toward higher frequency in the case of η_N. The PCA performed on the aggregated annual values showed that nutrient load fluctuations had a decoupling impact on the link between warming-induced changes in

R^* and nutrient uptake efficiency (Fig. 7). The decreasing trend under low-level, small-scale fluctuations and the increase under high nutrient loads with larger fluctuation did not show significant correlation with either the warming-induced dissimilarity or with the change in mean R^*. Changes in t_g, however, showed moderate positive relationships with both dissimilarity and changes in uptake efficiency (Fig. S4). The first two principal components explained 65% of total variance.

Best-fit GLS models, which included all temperature scenarios and nutrient load levels, showed more intricate relationships (Table IV). Net growth time was dropped from the models due to non-significant effects. Nutrient load level and temperature had strong positive effects, while R^* exhibited a significant negative impact on uptake efficiency. Temperature-related interactions were similar for both nutrients, showing a significant negative interaction with the effect of load level and a significant positive relationship with the impact of fluctuation.

Discussion

The simulation outputs imply that the direct impact of long-term warming on phytoplankton nutrient uptake in freshwater lakes can depend on the trait structure of the species pool as much as on fluctuations in major limiting nutrients. In general, higher elevated mean temperature led to higher annual nutrient uptake efficiency and to simultaneously more balanced nutrient uptake kinetics during the seasonal cycle. Higher uptake efficiency suggests that phytoplankton can retain a larger portion of nutrients entering the system over a year, which thus may also provide a more consistent flux of nutrients toward higher trophic levels. In cases when variability in response curves within the initial species pool promoted high compositional turnover under warming, the rise in uptake efficiency was constrained. This limitation was accompanied by an increase in community-weighted minimum resource requirements, which became an influential alteration in trait structure during the nutrient-limited summer period. However, with increasing fluctuations in nutrient load, the role of compositional shifts became weaker, though still significant.

Even though the link between inorganic nutrient concentrations and algal growth follows well-known Monod-type kinetics (Tilman et al., 1982), interspecific, temperature-dependent variability in functional response curves makes the estimation of future changes in phytoplankton-mediated ecosystem processes difficult (Barton & Yvon-Durocher, 2019). Therefore, incorporating the aspects of species dynamics, resource utilization traits, temperature patterns and various nutrient loads in a common framework is a prerequisite for improving estimation potential in future freshwater ecosystem responses to climate change. The presented model simulations contribute to this effort with several combinations of these factors.

Beyond global warming in general, the impacts of climate change also concern annual temperature patterns and their consequences on ecosystem processes. Global summer lake water temperatures were found to show an increasing tendency during recent decades (O'Reilly et al., 2015), and a similar trend was detected in the lake the present simulations were based on (Lake Balaton) (Pálffy & Vörös, 2019; Li et al., 2024). Increasing annual mean temperature also drove phenological shifts, causing an earlier onset of the growing season (Li et al. 2026), which is in agreement with the simulation outputs. Nutrient cycling and phytoplankton dynamics are inherently coupled and strongly influenced by temperature, thus, earlier spring blooms are also known to result in earlier onset of nutrient limitation (Meis et al., 2009). This modification in seasonality effects is an important point in the present study: a warming climate can result in more efficient phytoplankton nutrient uptake over the year, but the high-temperature, nutrient-limited period might cause shifts in trait structure that tend to temporarily reduce uptake efficiency. A key challenge in elucidating what these alterations can lead to in terms of ecosystem processes is to find essential associations between species presence and functioning in communities exposed to a constantly changing environment.

A critical point in interpreting the presented outcomes concerns what environmental context the simulation conditions represent. The fact that Lake Balaton as a large shallow lake has a regularly mixed water column makes a relatively simple model a suitable tool to study plankton dynamics in the lake. In the case of deeper, non-polymictic lakes, where stratification drives drastic shifts in physicochemical conditions and plankton composition along the water column, a similar study would presumably require a more complex, spatially explicit structure. In comparison, thermal stratification in Lake Balaton is occasional and short-term, which can nonetheless indirectly trigger temporary nutrient release from the sediment (Istvánovics et al., 2022), increasing seasonal variability in nutrient supply. This phenomenon is a clear example where the effects of warming and nutrient variability intertwine. As shallow lakes are highly sensitive to warming (Meerhoff et al., 2012), the presented simulations may be particularly relevant to these systems, calling for further mechanistic studies on climate change-driven alterations in phytoplankton-mediated nutrient fluxes.

Regarding nutrient supply and limitation, it is widely recognized that the ratio of N and P loads is as important as their magnitude (Hecky & Kilham, 1988). The effect of the N:P ratio was also apparent in the constant load simulation outcomes, i.e. warming-induced rises in uptake efficiency mostly occurred in the case of the limiting nutrient. Experimental evidence shows that temperature variability and nutrient supply interactively influence phytoplankton physiology and stoichiometry (Gerhard et al., 2019), although communities tested under laboratory conditions within a short time span may display responses that are difficult to extrapolate. When and whether N or P becomes

limiting in a specific system is rather varied, and co-limitation is overall more common, influenced by numerous factors, e.g. seasonality (Søndergaard et al., 2017) and depth (Sterner, 1994). This is further complicated by modifications in lake thermal structure, such as those observed in Lake Balaton, leading to occasional boosts in summer internal P load (Istvánovics et al., 2022). Regardless of how the fluctuations in supply alter nutrient ratios, the results of the present study suggest that higher temperatures support more effective and balanced uptake of nutrients, which may transfer to alterations in phytoplankton stoichiometry and also to indirect shifts in flux ratios within the planktonic food web.

Efforts to link traits to nutrient utilization have a long history. A crucial point in this respect is that the role of maximum growth rate and minimum resource requirements in competitive success within the phytoplankton highly depends on the variability in nutrient supply (Grover, 1990). There is also substantial evidence that species-specific resource requirements vary with temperature, which also influences competitive success (Lewington-Pearce et al., 2019). Large taxonomic groups of algae and cyanobacteria cover somewhat different niches along the multidimensional gradients of temperature and nutrients (Litchman, 2023), however, it is still difficult to assess the actual impact of these niche differences on phytoplankton dynamics and closely coupled elemental cycles. The presented constant load simulations show that if trait variability within the species pool triggered species replacement under elevated annual temperatures, it led to the dominance of species with higher minimum resource requirements, which inevitably constrained the warming-induced increase in nutrient uptake. This particular mechanistic link between structure and functioning pervaded most nutrient load levels and ratios. The apparent contradiction of higher R^* under a warming scenario despite the shift in dominance can be explained by the higher thermal optimum of the prevailing species and the species-specific temperature-dependence of R^*. This combined change in temperature- and nutrient-related traits ensured competitive success for these species over the ones dominating the baseline scenario with a concurrently limited increase in nutrient uptake efficiency. The observed contrast between warming-induced competitive success and constrained uptake warrants further investigation, as it emerges from the characteristics of the applied species response curves. Establishing whether this phenomenon is similarly common under field conditions would require substantially more research into intra- and interspecific variation in response curves across a range of algal groups.

External and internal nutrient sources in natural aquatic systems are diverse and generate spatiotemporally varying fluxes (Collins et al., 2017). That is why taking the influence of nutrient supply fluctuations into account is vital to fully understand the mechanisms behind phytoplankton dynamics. The effect of these fluctuations in the current study complemented the warming-induced impact of compositional shifts. Higher loads with higher variability generally led to enhanced

uptake efficiency in response to warming, whereas lower loads with moderate fluctuations tended to diminish uptake, independently of changes in community-weighted mean R^*. In addition to trait structure alterations, concomitant increases in phytoplankton net growth time could also be associated with the demonstrated increase in uptake, although its impact is not straightforward. This outcome raises an important question: is changing community-wide uptake kinetics *per se* or altered species composition a more defining driver of enhanced nutrient uptake under a warmer annual cycle? This subject requires more research in the future, since modifications in trait structure can take place in multiple ways. On top of compositional shifts, both temperature and the N:P ratio can drive intraspecific trait variability (Hofmann et al., 2019), and evolutionary changes in R^* can also improve species-level competitive abilities (Gallego & Narwani, 2022).

However crucial warming and nutrient loads are in long-term phytoplankton responses to climate change, the additional effect of concurrently altered lake conditions, including shrinking ice cover, decreasing water depth or modified mixing regimes (Woolway et al., 2020) need to be considered. These environmental shifts also affect light conditions, strongly interacting with the effect of temperature (Edwards et al., 2016), which is beyond the scope of this study. Despite these limitations, the presented simulations show that linking interspecific variability in functional response traits to nutrient fluxes can yield additional mechanistic insight and generate new hypotheses on how global warming can impact phytoplankton resource use. Fluctuations in the abiotic environment and community restructuring are both influential factors in phytoplankton-mediated ecosystem processes, and their relative importance may significantly shape future responses to long-term warming in lakes.

Conclusion

This modelling study demonstrated that the effects of long-term warming on phytoplankton nutrient uptake can be influenced by both the trait structure of the communities and by the ratio and temporal variation of nitrogen and phosphorus loads. Overall, higher annual mean temperature resulted in higher annual nutrient uptake efficiency during the simulations, and this increase was more pronounced in the case of the limiting nutrient. However, when the set of growth response curves within the species pool facilitated strong species turnover under warming, the rise in annual nutrient uptake was limited due to increase in community-weighted minimum resource requirements. On the other hand, when nutrient loads were set to fluctuate, increasing fluctuations enhanced the warming-induced increase in uptake efficiency, independently of the effect of shifting minimum resource requirements. The contrasting impacts of response trait structure and load

fluctuation suggest that the rate of compositional shifts and variations in the environment can have a decisive role in how phytoplankton functioning reacts to climate change in the long run.

Acknowledgements

This work was supported by the National Multidisciplinary Laboratory for Climate Change project (RRF-2.3.1-21-2022-00014). The author gratefully acknowledges the constructive comments of two anonymous reviewers, which helped improve the manuscript.

Funding

This study was funded by the National Research, Development and Innovation Office (RRF-2.3.1-21-2022-00014 Project).

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Figures

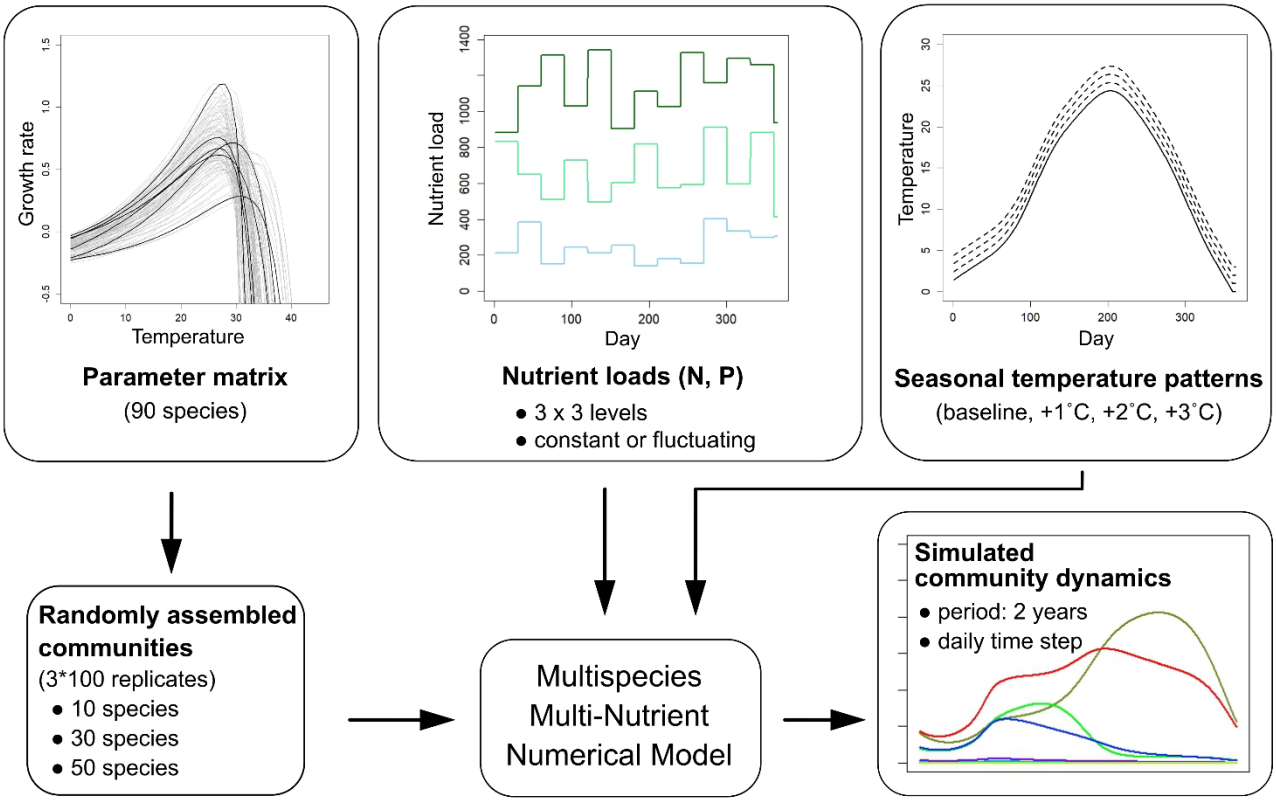


Fig. 1. Schematic representation of the model simulation environment.

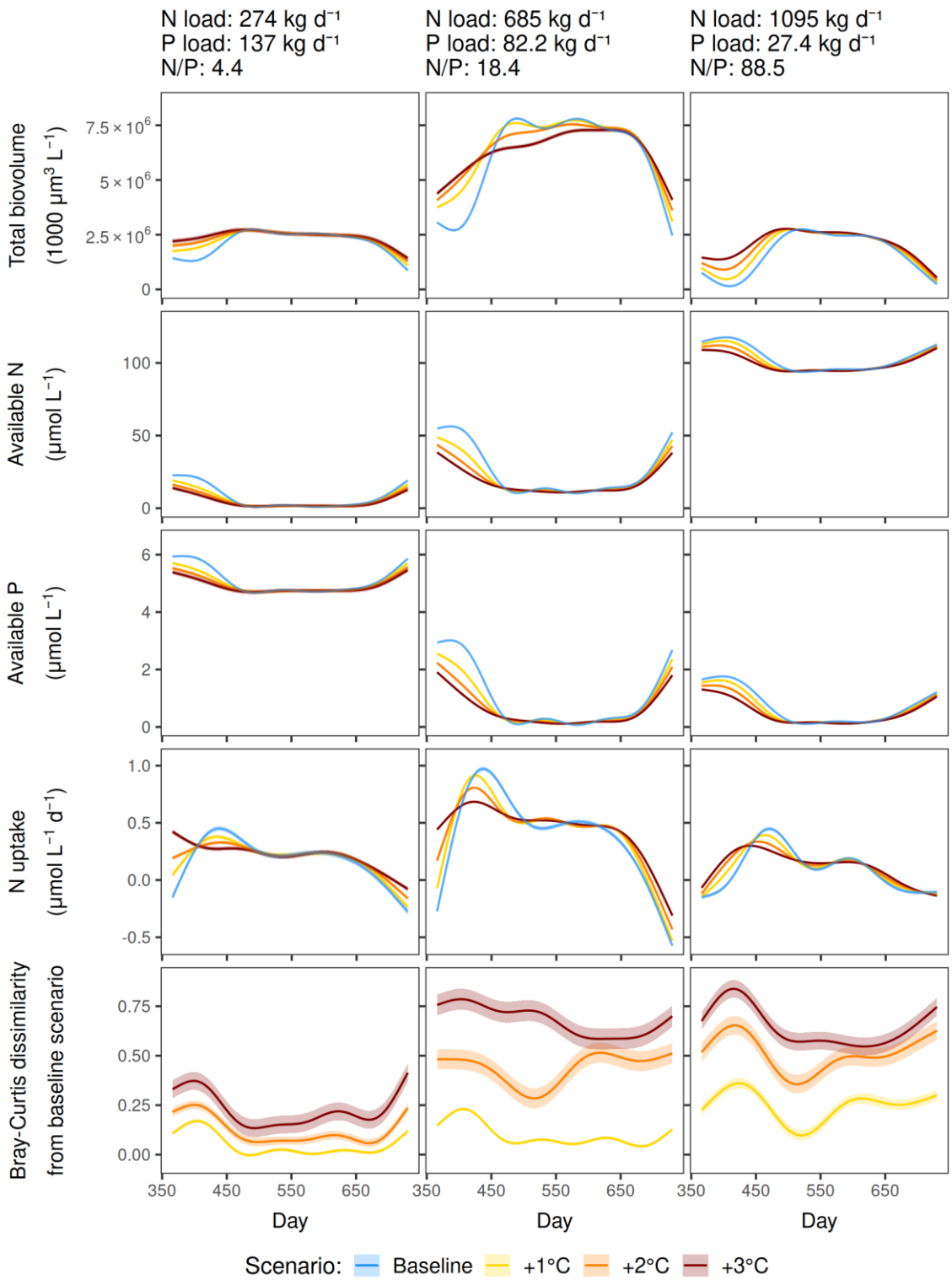


Fig. 2. Daily changes in phytoplankton-related metrics and nutrient concentrations under different temperature scenarios in simulations with constant nutrient supply. Metrics presented (starting from the top row): total phytoplankton biomass, concentrations of inorganic nitrogen (N) and phosphorus (P), nitrogen uptake, and Bray-Curtis dissimilarity from the baseline scenario. Negative values in N uptake represent mortality-associated N loss. Columns represent different molar N:P load ratios. Plots show the second annual cycle, lines represent GAMM (generalized additive mixed model) fits, shaded areas are 95% confidence intervals.

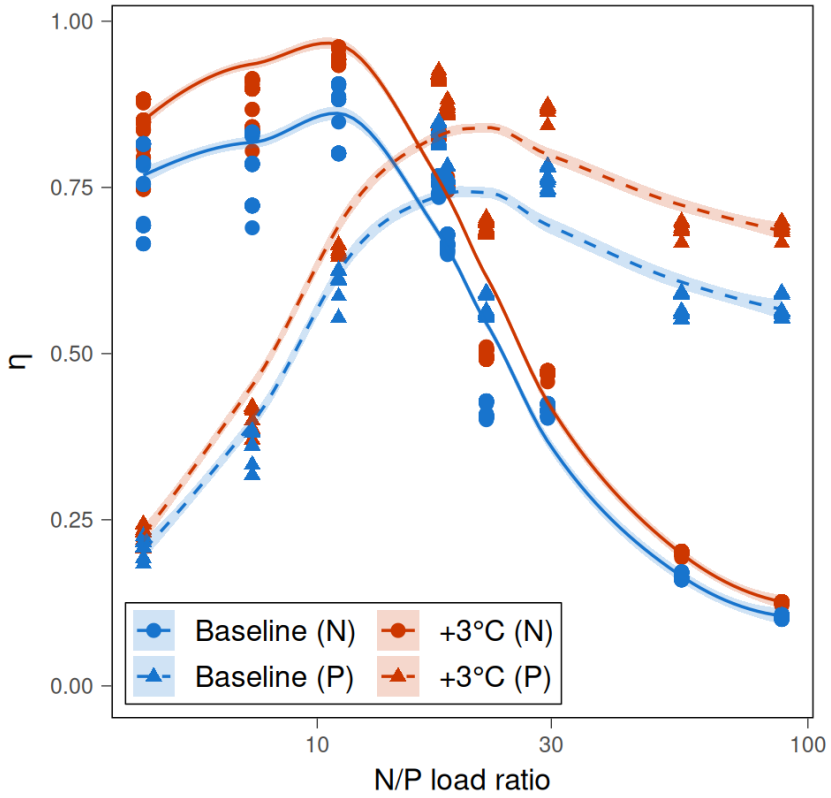


Fig. 3. Annual nitrogen (N) and phosphorus (P) uptake efficiencies (η) versus the molar N/P load ratio in baseline and +3°C temperature scenarios. Lines represent GAMM (generalized additive mixed model) fits, shaded areas are 95% confidence intervals. The plot shows simulations with communities of an initial species richness of 50, run at constant nutrient loads. The horizontal axis is log-scaled.

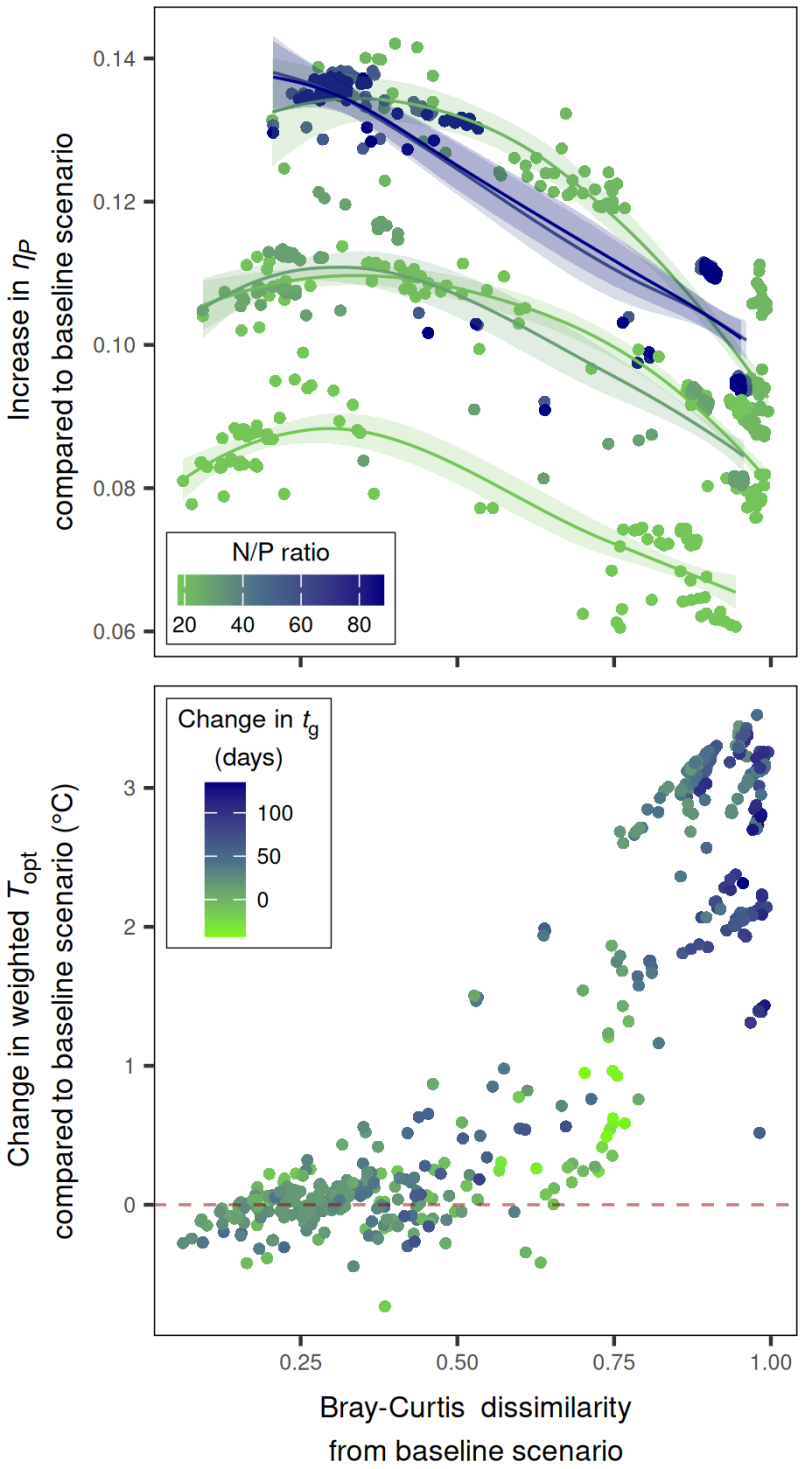


Fig. 4. Increase in annual phosphorus uptake efficiency (η_P, top figure) and changes in biovolume-weighted mean annual thermal optimum (T_{opt}, bottom figure) under +3°C warming versus mean annual Bray-Curtis dissimilarity from the baseline temperature scenario. Lines represent GAMM (generalized additive mixed model) fits, shaded areas are 95% confidence intervals. N/P ratio: nitrogen-to-phosphorus load ratio; t_g: annual net growth time. Both plots show simulations with communities of an initial species richness of 50, run at constant nutrient loads exceeding an N/P ratio of 16:1 to filter for P-limiting conditions.

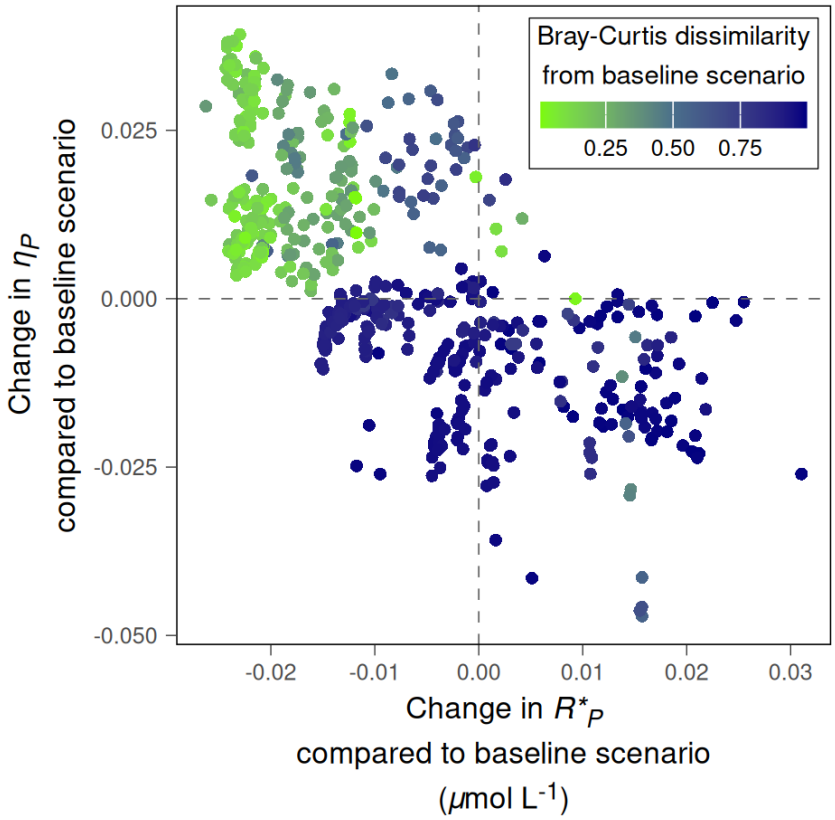


Fig. 5. Changes in phosphorus uptake efficiency (η_P) versus changes in biovolume-weighted mean minimum requirement for phosphorus (R^*_P) in response to +3°C warming compared with the baseline temperature scenario, during the period of nutrient limitation. The plot shows simulations with communities of an initial species richness of 50, run at constant nutrient loads exceeding an N/P load ratio of 16:1 to filter for P-limiting conditions.

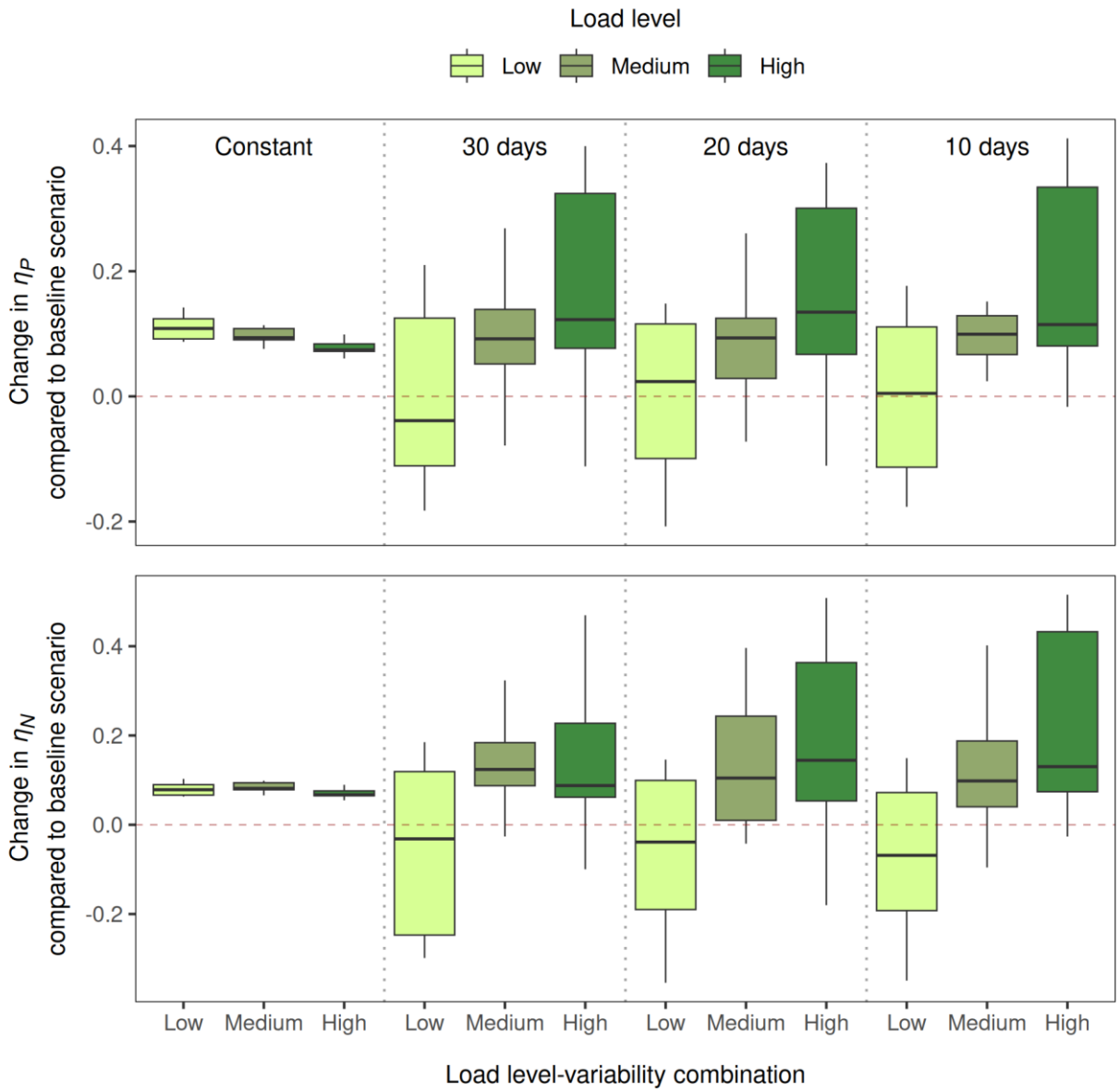


Fig. 6. Boxplots of changes in annual phosphorus and nitrogen uptake efficiency (η_P, η_N) in response to +3°C warming compared with the baseline temperature scenario under fluctuating nutrient load conditions. Subplots divided with dashed lines represent simulations with different rates of temporal variation in nutrient loads, increasing from left to right. The plot shows simulations with communities of an initial species richness of 50.

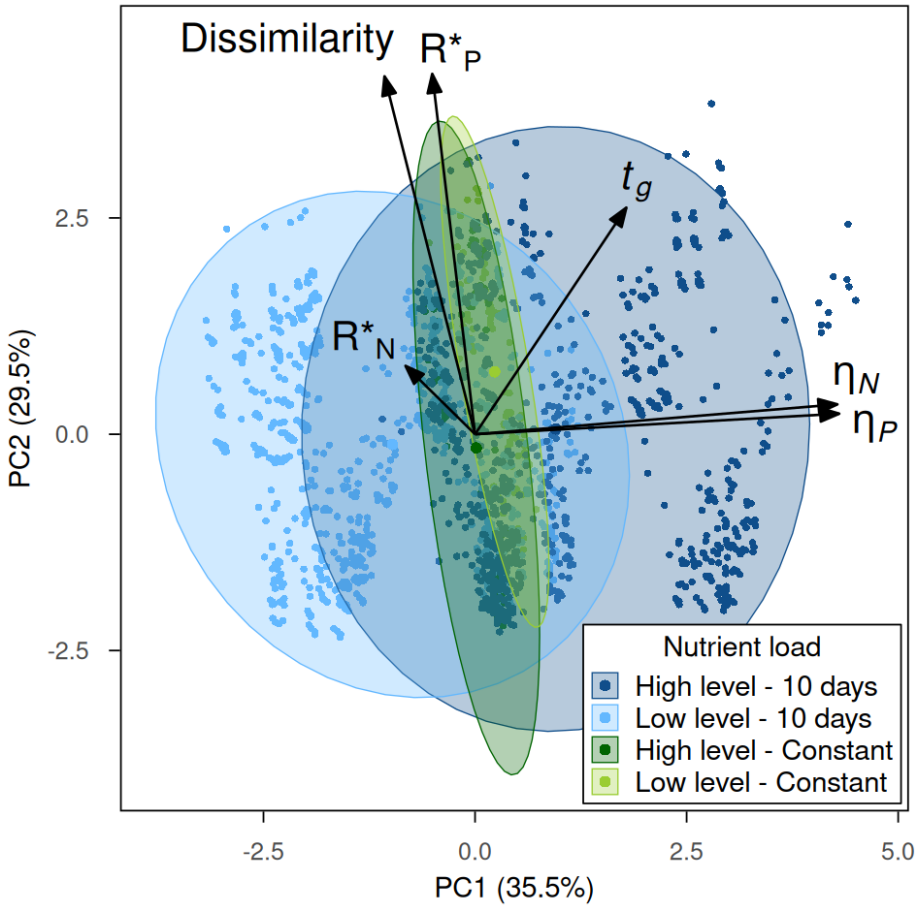


Fig. 7. PCA biplot of Bray-Curtis dissimilarity and changes in mean annual minimum requirements for nitrogen and phosphorus (R^*_N, R^*_P), in annual net growth time (t_g) and in nitrogen and phosphorus uptake efficiency (η_P, η_N) in response to +3°C warming compared with the baseline temperature scenario. The plot shows simulations with communities of an initial species richness of 50, run at constant nutrient load levels and at fluctuating loads with a frequency of 10 days. Shaded areas represent 95% probability ellipses.

Tables

Table I. Variables and model parameters.

Variable	Description	Unit
t	time step	day
T	temperature	°C
R_j	concentration of nutrient j	µmol L^{-1}
R_{Sj}	concentration of nutrient j in the source	µmol L^{-1}
N	concentration of inorganic nitrogen	µmol L^{-1}
P	concentration of inorganic phosphorus	µmol L^{-1}
A_i	abundance of species i	cell L^{-1}
Parameter	Description	Unit
μ_i	instantaneous growth rate of species i	day^{-1}
μ_{ij}	specific growth rate of species i at T and R_j	day^{-1}
b_{1ij}	theoretical growth rate at 0°C	day^{-1}
b_{2ij}	exponential change in growth rate with changing T	$°C^{-1}$
d_{0ij}	T-independent mortality term	day^{-1}
d_{1ij}	T-dependent mortality term	day^{-1}
d_{2ij}	exponential increase in mortality rate with T	$°C^{-1}$
K_{ij}	half-saturation constant of species i for nutrient j	µmol L^{-1}
q_{ij}	cellular quote of species i for nutrient j	µmol $cell^{-1}$
v	hydraulic flushing rate	day^{-1}
U_j	daily uptake for nutrient j	µmol L^{-1} day^{-1}
$\Delta R_{t,t+1}$	change in inorganic nutrient concentration within the system in time step t	µmol L^{-1} day^{-1}
η_j	uptake efficiency for nutrient j	-
R^*_{ij}	minimum requirement of species i for nutrient j	µmol L^{-1}
t_g	net growth time	day

Table II. Hydrological conditions and nutrient loads used in the model simulations.

Environment	Hydrological load (1000 m^3 day^{-1})	N load (kg day^{-1})	P load (kg day^{-1})
Constant loads	685	274	27.4
			82.2
			137.0
		685	27.4
			82.2
			137.0
		1095	27.4
			82.2
			137.0
Fluctuating loads (changing every 30, 20 or 10 days)	(548 – 822)	(137 – 411)	(13.7 – 41.1)
		(411 – 959)	(54.8 – 109.6)
		(822 – 1370)	(82.2 – 191.8)

Table III. Summary statistics of best-fit generalized additive mixed models (GAMM) fitted on annual N and P uptake efficiency (η_N and η_P) as response variables in constant nutrient load simulations. s(): smoothing spline; ti(): tensor product interaction; S: initial species richness; R^*_N, R^*_P: annual means of minimum nitrogen and phosphorus requirements for growth; t_g: net growth time; N/P: nitrogen-to-phosphorus load ratio; Edf: estimated degrees of freedom; AIC: Akaike's Information Criterion; n: sample number.

Response variable	Coefficients (explanatory variables)	Estimate	SE	t value	p value
η_N	Intercept	0.5708	0.0008	681.11	<0.0001
	Temperature (scenario)	0.0192	0.0008	23.00	<0.0001
	S	0.0025	0.0001	20.00	<0.0001
	R^*_N	0.0011	0.0001	10.68	<0.0001
	t_g	0.0146	0.0004	40.48	<0.0001
	Smooth terms	Edf		F value	p value
	s(N/P)	3.998		86816	<0.0001
	ti(N/P, Temperature)	5.339		169	<0.0001
	Variance structure:	power of covariate (varPower)			
	AIC:	-33962.52			
	Adjusted R^2:	0.92			
	n:	10800			
Response variable	Coefficients (explanatory variables)	Estimate	SE	t value	p value
η_P	Intercept	0.6336	0.0007	914.08	<0.0001
	Temperature (scenario)	0.0359	0.0008	44.07	<0.0001
	S	0,0109	0.0004	27.27	<0.0001
	R^*_P	-0.0050	0.0004	-13.42	<0.0001
	t_g	0.0094	0.0006	16.85	<0.0001
	Smooth terms	Edf		F value	p value
	s(N/P)	4.000		46919	<0.0001
	ti(N/P, Temperature)	8.802		152	<0.0001
	Variance structure:	constant plus power of covariate (varConstPower)			
	AIC:	-31082.61			
	Adjusted R^2:	0.948			
	n:	10800			

Table IV. Summary statistics of best-fit generalized least squares models (GLS) fitted on annual N and P uptake efficiency (η_N and η_P) as response variables in fluctuating nutrient load simulations, filtered for communities with an initial species richness of 50. Colons between explanatory variables indicate interaction terms. R^*_N, R^*_P: annual means of minimum nitrogen and phosphorus requirements for growth; AIC: Akaike's Information Criterion; RMSE: root mean squared error; n: sample number.

Response variable	Coefficients (explanatory variables)	Estimate	SE	t value	p value
η_N	Intercept	0.6538	0.0026	254.18	<0.0001
	Fluctuation	0.0062	0.0026	2.37	0.0177
	Load level	0.1606	0.0025	64.48	<0.0001
	Temperature (scenario)	0.0232	0.0023	9.91	<0.0001
	R^*_N	-0.0145	0.0005	-29.77	<0.0001
	Load level : Temperature	-0.0013	0.0004	-2.94	0.0033
	Fluctuation : Load level	-0.0061	0.0025	-2.38	0.0173
	Fluctuation : Temperature	0.0086	0.0024	3.61	0.0003
	Variance structure:	power of covariate (varPower)			
	AIC:	-78430.42			
	Residual standard deviation (σ):	0.1280			
	RMSE:	0.0883			
	n:	39600			
Response variable	Coefficients (explanatory variables)	Estimate	SE	t value	p value
η_P	Intercept	0.7880	0.0013	618.13	<0.0001
	Fluctuation	-0.0127	0.0013	-9.80	<0.0001
	Load level	0.0850	0.0012	72.11	<0.0001
	Temperature (scenario)	0.0339	0.0010	33.770	<0.0001
	R^*_P	-0.0041	0.0002	-22.83	<0.0001
	Load level : Temperature	-0.0050	0.0002	-21.59	<0.0001
	Fluctuation : Load level	0.0011	0.0012	0.95	0.3405
	Fluctuation : Temperature	0.0029	0.0010	2.86	0.0043
	Variance structure:	exponential of covariate (varExp)			
	AIC:	-131501.4			
	Residual standard deviation (σ):	1.2803			
	RMSE:	0.0469			
	n:	39600			

Supplementary material
for the manuscript

"Contrasting effects of compositional shifts and resource fluctuations on phytoplankton nutrient uptake under warming"

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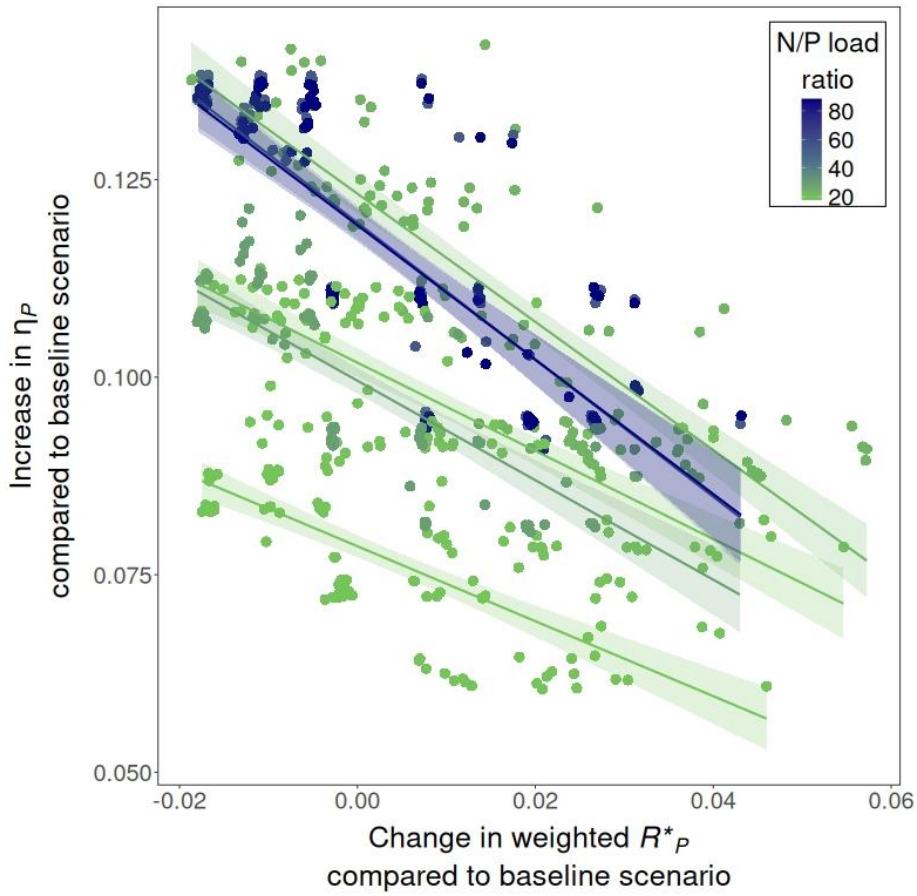


Fig. S1. Increase in annual phosphorus uptake efficiency (η_P) versus changes in biovolume-weighted mean annual minimum requirement for phosphorus (R^*_P) under +3°C warming compared with the baseline temperature scenario. The plot shows simulations with communities of an initial species richness of 50, run at constant nutrient loads exceeding an N/P ratio of 16:1 to filter for P-limiting conditions. Lines represent linear regression fits, shaded areas show 95% confidence intervals.

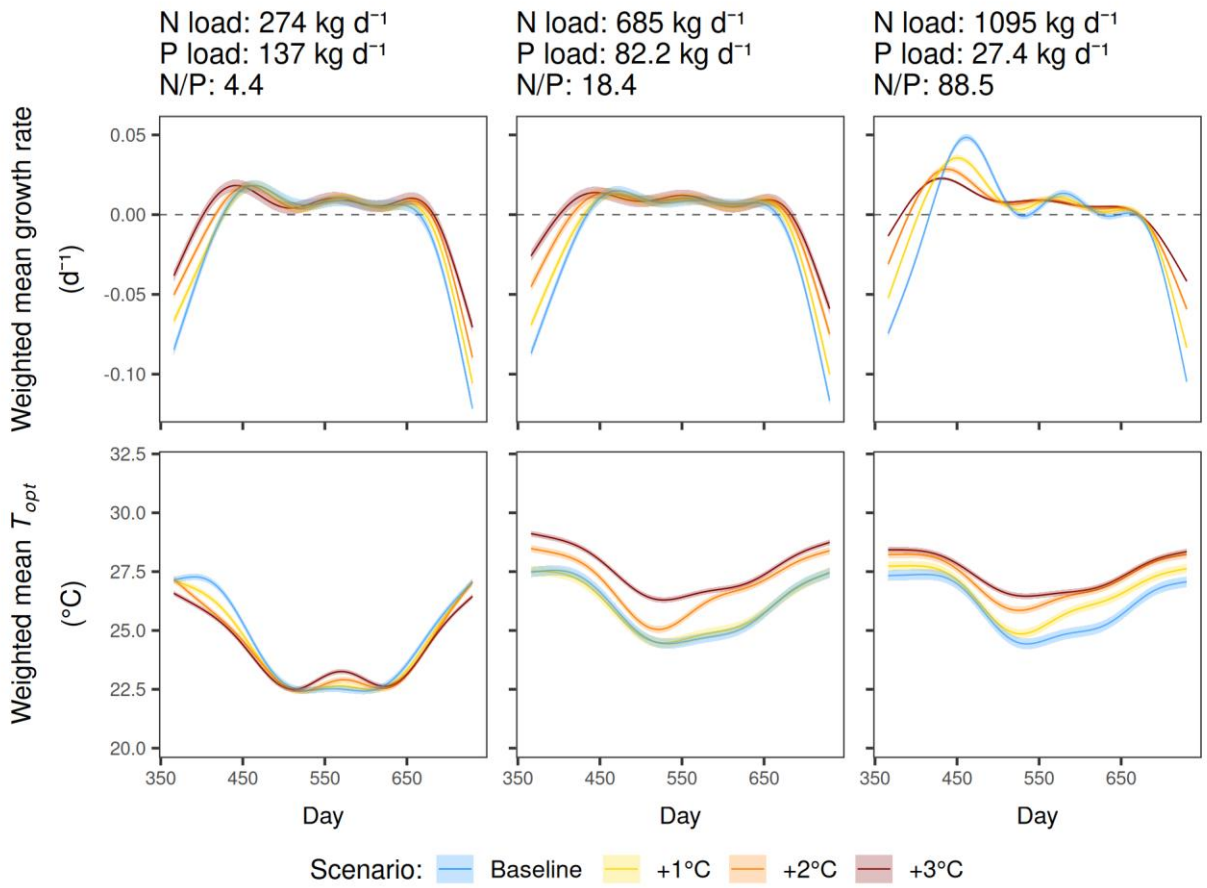


Fig. S2. Daily changes in biovolume-weighted mean growth rate and thermal optimum (T_{opt}) under different temperature scenarios in simulations with constant nutrient supply. Columns represent different N:P load ratios. Plots show the second annual cycle, lines represent GAMM (generalized additive mixed model) fits, shaded areas are 95% confidence intervals.

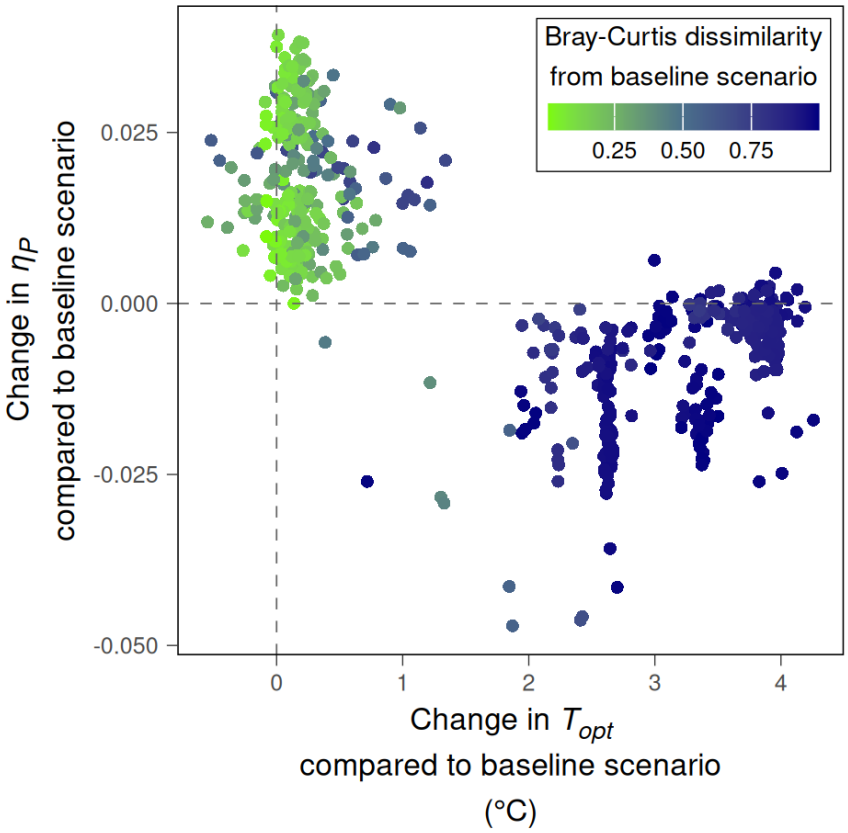


Fig. S3. Changes in phosphorus uptake efficiency (η_P) versus changes in biovolume-weighted mean thermal optimum (T_{opt}) in response to +3°C warming compared with the baseline temperature scenario, during the period of nutrient limitation. The plot shows simulations with communities of an initial species richness of 50, run at constant nutrient loads exceeding an N/P load ratio of 16:1 to filter for P-limiting conditions.

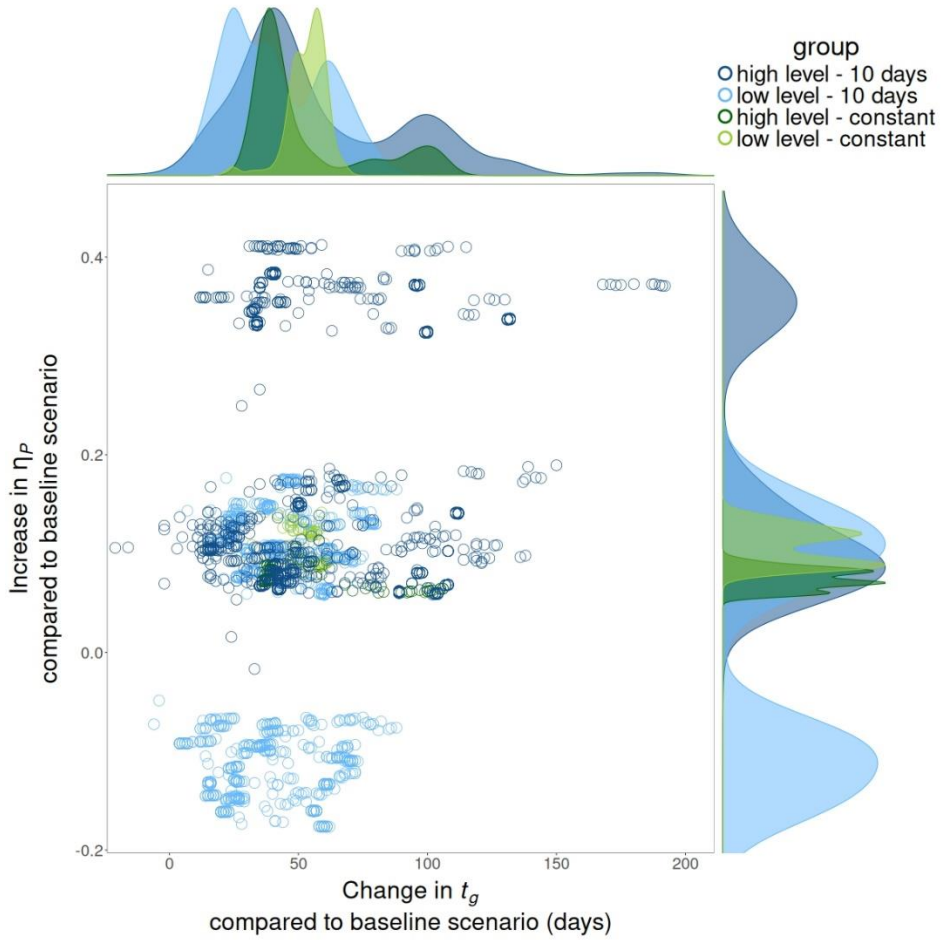


Fig. S4. Increase in annual phosphorus uptake efficiency (η_P) versus changes in annual net growth time (t_g). The plot shows simulations with communities of an initial species richness of 50, run at constant nutrient load levels and at fluctuating loads with a frequency of 10 days. Subplots on top and on the right show normalized density plots of the respective variables in the indicated load level-fluctuation groups.